



In Unity There Is Strength: How to Incorporate Your Research into Teaching

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INTRODUCTION

Often time when academics talk about incorporating research in teaching, we refer to either training PhD students to do their original work or including students in your research as assistants or collaborators. However, most degree-granting postsecondary institutions in the United States are not research universities with high or very high levels of research activities,¹ and most academics' jobs do not focus on training PhD students. When teaching involves professional-track master students or undergraduates, our job centers more on helping students learn about a subject systematically. As such, our research and teaching become more disconnected and go their separate ways. Integrating research and

¹ https://nces.ed.gov/programs/digest/d20/tables/dt20_317.20.asp (Accessed on June 21, 2022).

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teaching activities becomes increasingly challenging when teaching undergraduate or professional-track master students who have limited exposure to conducting original research.

This chapter is about how to actively and effectively integrate our research and teaching for academics teaching non-PhD students. There are three immediate benefits in closing the gap between our research and teaching. First, we already spend a lot of time and energy thinking about our research, which is what we are passionate about. Finding ways to connect our research and teaching allows us to channel the passion and connect it with our students. Students get motivated when they feel our enthusiasm for the subject, and motivated students achieve better learning outcomes (Ambrose et al., 2010). Second, embedding our research in our teaching also gives us a way to approach our research differently, and sometimes it helps improve our research, especially in terms of how to frame and communicate our research with a broader audience. Last, in the long run, it simply saves us time and mental energy to transit between research and teaching.

When teaching non-PhD students, it requires more conscious decisions and course designs to connect research and teaching, and this is what this chapter is about. This chapter examines a variety of ways our research can facilitate our teaching and offers tips to identify opportunities in our research process that can be turned into teaching moments to improve student engagement. To this end, the chapter is divided into two parts. The first part identifies teaching opportunities in the research process. I organize different strategies based on the level of preparation or adaptation needed in the process of turning those research steps into teaching opportunities. The second part provides concrete tips on how to actively embed your research in your teaching. This section can be particularly useful for instructors at Primarily Undergraduate Institutions (PUIs) where research resources are more limited and teaching and service loadings can be more demanding. I use one introductory undergraduate course as a running example to demonstrate concrete steps of how to actively incorporate research into undergraduate teaching. I conclude the chapter with a brief discussion on how your research can motivate your students toward more effective learning.

Because of my experiences working in a PUI setting, this chapter is written with three types of academics in mind. It would certainly be most useful for instructors working in the PUI setting. Besides, the chapter may also be useful for novice instructors or teacher-scholars who are trying to

integrate their teaching and research more. My overall goal is to help academics with more demanding teaching obligations to find ways to channel their research passion into more effective teaching.

TEACHING OPPORTUNITIES IN THE RESEARCH PROCESS

How to identify teachable moments in your research process? In this section, I list some steps in our research process that can become good teaching opportunities. I organize the strategies based on the needed adaptation efforts. I start with the strategy with a minimum level of preparation to the one with a maximum level.

Teach What You Want to Read for Your Research

The most straightforward way is to assign readings that you are going to read for your research. This strategy requires almost zero to minimum preparation and is most suitable for undergraduate upper-level seminars or graduate-level courses. This strategy is particularly useful for researchers who are in the process of writing a book manuscript. The literature review section in a book manuscript is usually more extensive regarding the breadth and depth of the works reviewed. Each strand of research can easily become a week's reading list. The literature review section can easily become the skeleton schedule for your syllabus. You can also use teaching as an opportunity to help you organize and summarize your understanding of a research topic or your literature review section. For instance, you can assign what you plan to read to your students and have them summarize and evaluate the literature. During seminar discussions, you may use the opportunity to deepen your thoughts and analyses of the literature. If you happen to be writing the book when you are teaching the course, the benefit can be immediate and multiplying.

Use the Theoretical Debates in Your Research

Another useful strategy is to turn the conceptual/theoretical debates happening in your research area into an assignment or a discussion/debate topic in the classroom. Compared to the previous strategy, it takes more preparation to turn the theoretical debates into teaching opportunities and the focus of preparation is on how to tailor the theoretical discussion to fit the level of your course, which will be discussed more

extensively in the next section. The gist of this strategy is to use existing debates in the field and turn them directly into assignments or class activities.

For example, in comparative politics, one major debate is about what democracy is and is not and how to measure democracy. Another major debate is about what causes and sustains democracy. In my undergraduate-level research method class, I directly turn the debate on different conceptualizations of democracy into a series of assignments. I first have students compare and contrast how major comparative datasets (e.g., Freedom House, POLITY IV, V-Dem, etc.) conceptualize and operationalize democracy. I then have them use varied democracy proxies to study one country's democratization process.² In the research method class, students are then asked to use the case study to evaluate pros and cons of different conceptualizations. If it is a comparative politics course, I then ask students to apply theories of democratization and evaluate which argument explains the case better.

The “Background Learning” Strategy

When academics start a new research topic, we often need time to do some preliminary research. The preliminary research can of itself become teaching materials. The “background learning” strategy is useful especially when one is trying to start a new project/topic. Teaching obligations can motivate (or force) you to put in the time to do the initial research with concrete deadlines. Because it is also a new topic for you, the amount of preparation time needed can vary quite a lot. Depending on the course level and your energy level, there are two routes to adopt this strategy. First, one can choose the lecture route (which definitely requires more preparation). For instance, when COVID-19 started in 2020, I was curious about how regime types and state capacity impacted government variations in their COVID-19 responses. I did preliminary research and made it a lecture in my Asian Politics course, which students also highly appreciated given the timeliness of the topic was. The lecture also helped me organize my initial thoughts which ultimately led to several related publications (Yen, 2020; Yen & Liu, 2022; Yen et al., 2022). The second

² This part varies from class to class and depends on my research focus at the time. I usually ask students to focus on specific countries I want to know more about or keep track of.

route one can choose is to directly turn background information research into assignments for students. The caveat is that quality may vary quite a bit depending on students' levels. If you are to turn background information research into an assignment, I suggest two additional points: (1) still do some preliminary research yourself and point students toward the direction you want. For example, you can provide resources they should read and research; (2) limit the research scope for students. For instance, you may have different group's research different aspects of the background. In a nutshell, when you need more background information for your research, it can become a teaching opportunity as well.

Leveraging Community Engagement and Empirical Evidence

When it comes to finding teaching opportunities in research, what counts as research also matters. In many academic institutions (especially PUIs), scholarly activities are counted more broadly and there are research opportunities through community outreach. If community outreach is something you do, there are multiple ways in which we can combine "researching our community" with "teaching about our community." For instance, you can directly invite community partners to your classes, which requires minimum preparation. Another strategy is to have students analyze the data you collected (either qualitative or quantitative). For example, if you are teaching political participation and working on civic engagement in the community, it would be a great opportunity to have students look at the data and summarize what drives civic engagement in the neighborhood. In general, students get more motivated when researching the environment surrounding them. More broadly, you can modify the empirical data for your research, tailor them to fit the course levels, and have students summarize descriptive statistics and identify trends. Nevertheless, it may take longer than you think to prepare your empirical data for your students, so adopt this strategy only when you see a great fit between your empirical data and what you are teaching.

HOW TO INTEGRATE RESEARCH INTO TEACHING

The previous section touches on where and how your research process can inform your teaching. This section addresses the same topic in a slightly different fashion: if you already know what you are teaching, how can you actively incorporate more of your research into your teaching? Below

I provide concrete step-by-step tips to guide you through the process. To illustrate the step-by-step tips more vividly, I use one course I am currently teaching as a running example. The course is *Introduction to Comparative Politics*.

In a nutshell, a more productive process is to employ the strategy of backward design. That is, to start with the expected learning outcomes and the cognitive process students may have to go through to achieve the learning goals. Then, work backward to think through the assessment tools you want to use to evaluate students' success. Assessment tool design is a good place where you can actively incorporate your research. The next step is to organize the content into weekly lectures, discussion questions, and class activities. This is the second place where you can use your research to help you design your course content.

Why is it a more productive process to adopt a backward design approach? Because there are some fundamental differences between research and teaching. First, the end goal is different. Research is to produce systematic knowledge to enhance our understanding of the world around us. It is to use what we already know to explain what we do not know. As such, the end goal of research is to push for an explanation of the new phenomenon or the unknown. In contrast, teaching is to organize and "translate" the knowledge we already know so that students can understand the world around us through analytical frameworks, and this is especially true when teaching undergraduates, so the end goal is knowledge dissemination. Second, the audience is different. The audience of research is mostly peers in the field, who already have deep background knowledge of the subject. Students are the audience of your teaching. The further away students are from conducting original research, the bigger the gap is between research and teaching, and the more adaptation you would need from research to teaching. Due to the differences in the end goal as well as the audience, research should be tailored and incorporated differently depending on the levels of a course. Therefore, it is more productive to start from the endpoint, which is the learning objectives upon completion of a course, and design backward.

Step 1: Identify and Align Learning Objectives with Learning Typologies

To incorporate your research into teaching more effectively, the first and foremost is to identify what the learning objectives are. The learning

outcomes you set have direct implications on what and how your research should be incorporated. When identifying learning objectives, try using specific and measurable outcomes. Bloom's taxonomy provides a useful starting point. There are six levels of cognitive process in learning, including knowledge, comprehension, application, analysis, synthesis, and evaluation.³ Each of these levels represents one essential aspect of the learning process. Try to match the dimensions of the learning process with the objectives. Needless to say, your learning objectives should be closely related to the course levels.

Example: Introduction to Comparative Politics

Because it is an introductory course for the major as well as a general education course, the main objective is to lay the foundation for advanced courses and motivate undeclared students to get interested in political science. As a result, I focus on the cognitive domains of remembering and understanding fundamental political science concepts and the ability to apply these concepts in different contexts. My other goal is to introduce students to how social scientists study and measure political institutions. To this end, the learning objectives are (I bolded the verbs related to Bloom's learning taxonomy):

- **Recognize** and **explain** how political institutions at different levels affect people's lives and can **apply the knowledge** to different contexts and to real-world events.
- **Gain familiarity with** the resources available (i.e., **know where to find what data**) for studying politics in the developing world.

Step 2: Incorporate Your Research into Assessment Tools

Once you identify the learning objectives for a course and match the learning objectives with Bloom's learning typology, the second step is to think thoroughly about the assessment tools you want to use to evaluate your students' progress. This is where you can include some of the teaching opportunities discussed in the previous section. Background learning and theoretical debates are usually two ways I incorporate my research into assessment tools.

³ <https://bloomstaxonomy.net/> (Accessed on June 21, 2022).

Example: Introduction to Comparative Politics

To achieve the above-mentioned learning objectives, I need to design assessment tools to evaluate whether students can define and explain political concepts and apply those concepts in various contexts. Meanwhile, I also aim to familiarize students with existing datasets political scientists use to study the world.

Last time when I re-designed the course, one of my research projects was to study how state capacity shapes governmental variation in crisis response (Yen, et al., 2022). I turned to the dataset I was using at the time, the Quality of Government (QoG^{4,5}). Students then were asked to compare the two measures and how political scientists develop two different measures to study state capacity. Because I was already comparing different measures of state capacity for my research, I did not do much extra work to turn it into an assignment.

Step 3: Incorporate Research into Course Content

After you plan your assessment tools, the next step is to plan weekly lectures and class activities. Here, again, how your research can contribute to your course content is contingent upon the course level. If you are teaching an upper-level seminar, assigning students what you intend to read for your research would be a good strategy. If you are teaching a lower-level course, applying background learning to your research can be more useful. Sometimes, I also turn theoretical debates in the field into classroom debates.

Example: Introduction to Comparative Politics

For this introductory level course, I teach basic concepts such as modern state, regime types, and regime change. Given that there is constant development in regime change around the world, I always pick a new case to focus on when I teach regime change. For instance, in 2019, I wanted to keep track of Venezuela's presidential crisis. I turned the event first into my lecture material and then assignment content. Another example would be the 2021 regime change in Myanmar. I adopted the same strategy to stay informed about cases of democratic backsliding. On ongoing world

⁴ <https://qog.pol.gu.se/data/data-portal> (Accessed June 8th 2022).

⁵ <https://fragilestatesindex.org/>, (Accessed June 8th 2022).

events, Monkey Cage at The Washington Post often has good background readings for undergraduate students too, which makes teaching real-time events easier.⁶

MOTIVATION: THE SECRET SOURCE

Behind all the tips and strategies in better aligning our teaching with our ongoing research lies another fundamental factor for more effective learning: motivation. Research has shown that students' motivation has a direct impact on their learning behaviors and learning outcomes (Ambrose et al., 2010). Authentic and real-world applications can motivate students, and political science research often can be directly linked to ongoing real-world events. We political scientists are also often motivated by ongoing world events to continue our research. Try to channel what motivates you to do your research to your students. Help connect the dots for your students between the ongoing issues happening in the world and how our theoretical framework has relevance and offers useful perspectives. Your students will get motivated too. Moreover, we all have to pitch our research to the general audience, finding the right way to motivate students provides an opportunity for us as researchers to approach our research topics from different angles. The iterative process often helps us as researchers to frame our research better. In sum, higher levels of integration help channel one's passion for research into teaching and motivate students, which can become reciprocal stimulation for your ongoing research.

In a time when faculty burnout is a real problem in higher education (Gewin, 2021), it is of vital importance to find ways to work smarter but not harder to have a fulfilling career in academia. In this article, I discuss various ways research can inform teaching in political science. I identify opportunities in your research that can be transformed into teaching moments. The chapter also offers concrete tips on finding ways to incorporate research into your teaching. The primary suggestion is to start from the level of cognitive understanding we hope our students can achieve in the course before deciding how to incorporate research into

⁶ <https://www.washingtonpost.com/monkey-cage/> (Accessed June 8th 2022).

teaching. Once the learning objectives have been identified, work backward to pick and choose which part of your research can be tailored and included in teaching.

This chapter provides a skeleton structure and tips on how to actively incorporate your research into your teaching because the benefit is immense. Even though the suggestions are by no means exhaustive and need to be tailored according to the specific academic setting one is in, the chapter provides a starting point for academics working in teaching-oriented institutions to think about ways to bring their research closer to their teaching.

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